COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010: Data Structures and Algorithm	**SECTION:**	CPE21S3
DATE PERFORMED:	10 / 13 / 26	**DATE SUBMITTED:**	10 / 15 / 26
NAME/S:	Bautista, Troy Jerron A. Fabonan, Kurt Lorenz C. Geronimo, Joncel Zairho G. Salcedo, Lance Cristopher S. Santos, Paterno Miguel R.	**INSTRUCTOR:**	Engr. Jimlord Quejado

Seatwork No #11.1 Planning

1. PROCEDURE OUTPUT

Our 3 proposals:

1. **Automated Electric Vehicle Smart Charging & Grid Dispatch Hub:** Emulates a commercial EV charging station hub balancing limited power grid capacity, battery charge priorities, reservation slots, and billing transactions.
2. **Automated Emergency Dispatch & Hospital Bed Allocation System:** A smart city logistics simulation system designed to route 911 emergency response vehicles to incoming incident locations while dynamically matching patient severity to nearby hospital capacities and specialized equipment.
3. **Dynamic Cargo Express Freight & Multi-Modal Sorting Engine:** Emulates an automated courier logistics hub that manages incoming package sorting, warehouse storage allocation, vehicle load optimization, and real-time package delivery tracking.

Task	Lead responsibility	WEEK 1	WEEK 2	WEEK 3
Define the system's purpose, requirements, and project scope	Santos	■		
Choose the data structures and algorithm suited to the system	Salcedo	■		
Develop the main system features algorithm	Santos		■	
Connect the interface, data, and main features	Bautista, Fabonan		■	

Test the system and fix issues	Geronimo, Santos,Fabonan		■	
Prepare the report, presentation, and demonstration	Geronimo, Salcedo, Bautista			■

10 Questions about the Automated Electric Vehicle Smart Charging & Grid Dispatch Hub

1. How will the system prioritize incoming EVs when all charging bays are currently occupied?
2. What data structure will be used to track customer account details, billing history, and active session tokens?
3. How does the grid dispatch engine handle real-time power grid limits during peak load hours?
4. How are scheduled charging slot reservations handled to prevent double-booking?
5. How will the system organize dynamic pricing updates based on electricity tariff rates throughout the day?
6. What logic ensures fair queuing so low-priority EVs do not experience indefinite starvation?
7. How is the physical status of charging bays (Occupied, Available, Out-of-Service) maintained in memory?
8. How will the system compute optimal charging duration based on a user's target battery percentage and available power?
9. How does the system maintain transaction logs for audit trails and performance reporting?
10. How will user inputs and administrative overrides (e.g., emergency grid cutoff) be handled asynchronously in the user interface workflow?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

N/A

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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